

FunC-ZIDM: A Functional Concurrent Zero-Inflated Dirichlet Multinomial Model for Microbiome Analysis

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Joint work with Ander Wilson, Matt Koslovsky

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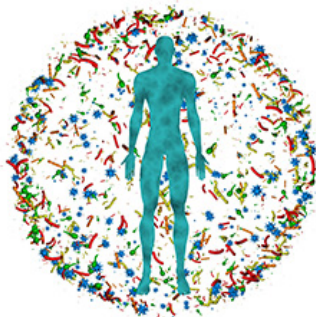
Human Microbiome Research

Human Microbiome: Community of microorganisms – bacteria, achaea, fungi, and viruses.

- The microbial community has been linked to digestive health, disease, and mental health.

Common goals are to understand how:

- the microbial community interacts with the host.
- exposures or treatments influence the microbial composition.
- the microbial composition changes over time.



The infant microbiome is of particular interest due to it's association with health and disease throughout life.

Longitudinal Microbiome Data Structure

- The sample times are often unbalanced and unequally spaced.
- The processed data are mapped to a reference database, which can be mapped at different levels of the phylogenetic tree.
- The sequencing of the sample yields a fixed number of reads.

Infant	Day	Taxa			
		1	2	...	J
1	1	242	0	...	841
1	4	200	36	...	649
1	15	50	156	...	403
2	6	0	0	...	946
⋮	⋮	⋮	⋮	⋮	⋮
N	36	491	0	...	0

Analysis Challenges

- Compositional structure.
- Overdispersion.
- High-dimensionality.
- Zero-inflation.
- Repeated measures and potentially functional outcomes and covariates.
- Potentially varying effects.

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Existing Methods

Challenge	Method			
	Count Models [5, 14–16]	Autoregressive [13]	TV Count Models [9, 12]	ZIDM [6]
Compositional	✗	✗	✗	✓
Overdispersion	✓	✗	✓	✓
High-dimensionality	✓	✓	✗	✓
Zero-inflation	✓	✗	✗	✓
Repeated measures	✓	✓	✓	✗
Functional	✗	✗	✓	✗

Functional Concurrent Zero-Inflated Dirichlet-Multinomial

$$\mathbf{z}_i(t) \sim \text{Multinomial}[\dot{z}_i(t), \boldsymbol{\psi}_i(t)], \quad \dot{z}_i(t) = \sum_{j=1}^J z_{ij}(t)$$

$$c_{ij}(t) | \eta_{ij} \sim (1 - \eta_{ij}) \delta_0(\cdot) + \eta_{ij} \text{Gamma}[\gamma_{ij}(t), 1]$$

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$1 \times J$ vector of taxa counts


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We regress onto the concentration parameters with a log-linear link function:

$$\log[\gamma_{ij}(t)] = \beta_{j0}(t) + \sum_{p=1}^P \beta_{jp}(t) x_{ip}(t) + r_{ij}.$$

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Time-varying coefficients

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- B-splines
- Regularized Horseshoe shrinkage [11].

Inference

- Multiplicative change in relative abundance:

$$\Delta_v \text{RA}_{jp} [t, \mathbf{x}(t)] = \exp [v\beta_{jp}(t)] \frac{\sum_{k=1}^J \exp [\mathbf{x}(t)\boldsymbol{\beta}_k(t)]}{\sum_{k=1}^J \exp [\mathbf{x}(t)\boldsymbol{\beta}_k(t) + v\beta_{kp}(t)]}.$$

- Multiplicative change in α -diversity:

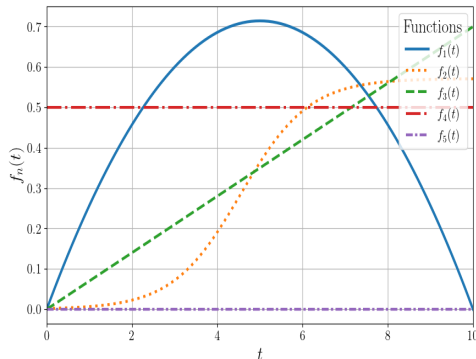
$$\Delta_v \alpha_p [t, \mathbf{x}(t)] = \frac{\alpha [t, \mathbf{x}^+(t)]}{\alpha [t, \mathbf{x}(t)]},$$

$\alpha [t, \mathbf{x}(t)] = \left[\sum_{j=1}^J \text{RA}_j [t, \mathbf{x}(t)] \{ \text{RA}_j [t, \mathbf{x}(t)] \}^{-l} \right]^{1/l}$ and $\mathbf{x}^+(t)$ is the covariate profile with a v -unit increase in covariate p .

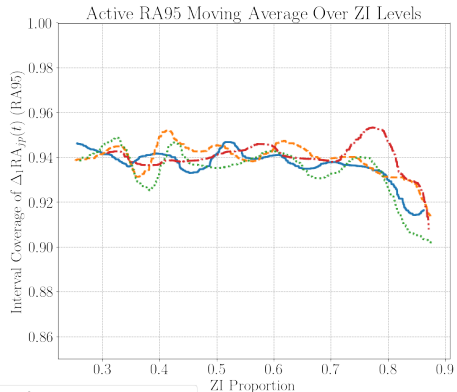
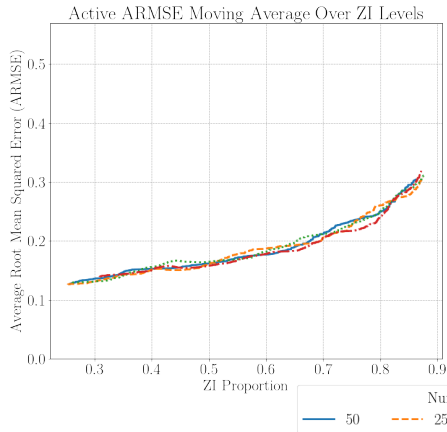
Simulation Setup

We explore the scalability of FunC-ZIDM for taxa size. We generate 100 datasets with:

- number of taxa being 50, 250, 500, or 1000,
- 50 individuals,
- 3 to 10 observations per individual,
- 10 covariates for each taxa,
- Minimum of 5 individuals with a non-zero count per taxa.



Simulation Results



- We also compare to similar models that remove the zero-inflation and/or random effects structure from our proposed model.

Case Study

We explore longitudinal infant microbiome data provided in La Rosa [7].

Samples:

- Fecal samples taken from 58 preterm infants between 1 to 80 days after birth.
- Number of samples range from 3 to 41, with a median of 15 and total of 922 samples.

Our objectives:

- How the relative abundance of each taxon changes over time.
- How covariates impact the relative abundances.
- How covariates impact overall microbiome diversity.

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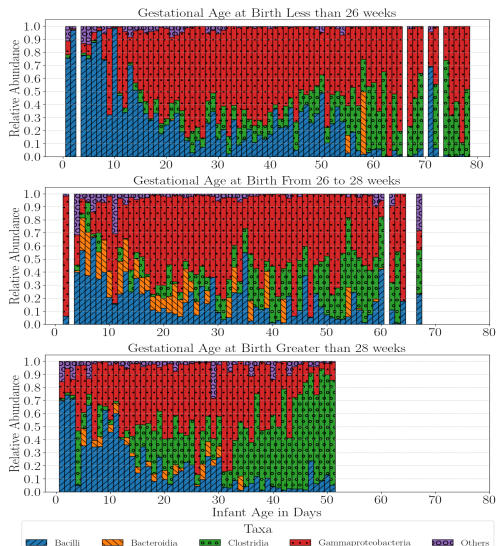
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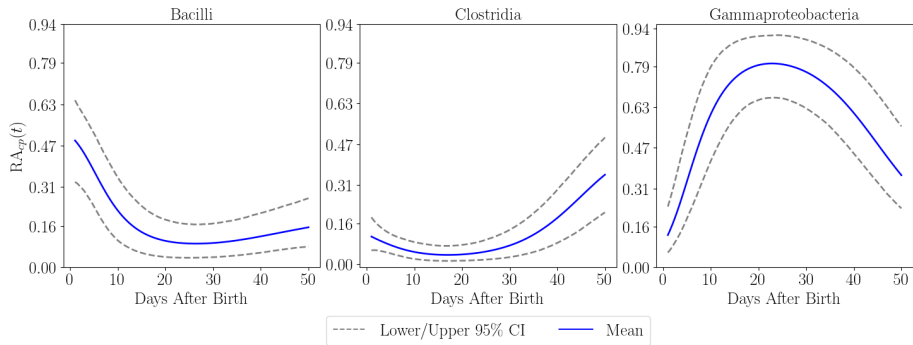
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Infant Microbiome Data

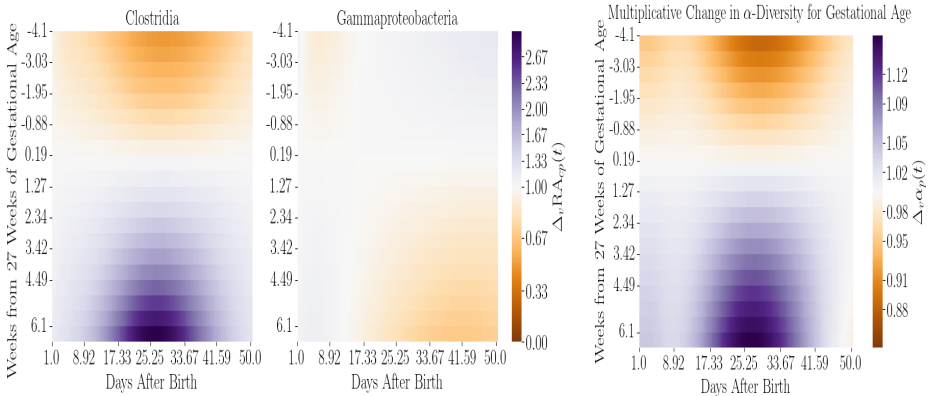
- Only taxa with ≥ 5 infants with ≥ 1 non-zero counts (16 taxa w/ max of 98.6% zeros).
- Only at-risk indicators probabilities prior specifications are not weakly informative.
- Covariates include breast milk in diet ($< 10\%$, $10 - 50\%$, and $> 50\%$) and gestational age at birth, amongst others.



Case Study Results: Baseline Relative Abundance

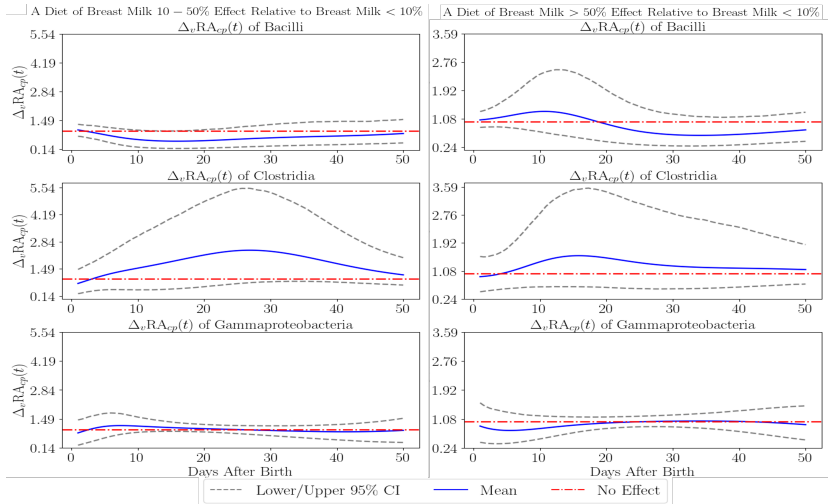


Case Study Results: Gestational Age After Birth



- Clostridia seems to be positively associated with gestational age over the whole time-period, whereas Gammaproteobacteria is mostly negatively associated.
- We also find the diversity increases as the infant is closer to full-term.

Case Study Results: Breast Milk Proportion



- We also find the diversity increases after about 30 days for a diet of breast milk of > 50% as compared to < 10%.

Case Study Results: Previous Studies

- Chernikova et al. [2] and Healy et al. [4] reported a lower microbial diversity for preterm infants compared to full term infants.
- Healy et al. [4] highlighted the dominance of Gammaproteobacteria in preterm infants controlling for breast milk, mode of delivery, and antibiotics.
- For breast fed infants, Ma et al. [8] documented a no change in α -diversity for the first 3 months of life, followed by significant change in 6 months of age.
- Ma et al. [8] and Gregory et al. [3] identified Bacilli as more prevalent in formula fed infants, but Pannaraj et al. [10] associated a lowering of Bacilli.

Conclusion

- Longitudinal data is common within human microbiome research.
- Current methods do not handle the compositionality, potentially leading to false positives, nor varying effects.
- We proposed FunC-ZIDM to handle the longitudinal and potentially functional effects, while accounting for compositionality and zero-inflation.
- Our simulation shows the method is scalable to a high amount of taxa.
- In the case study, we show similar results to pre-existing studies.
- Results are not consistent across all studies, which could be due to not accounting for compositionality or averaging effects over the whole time period.

Thank You!

- Website: <https://brodyee.github.io/>
- Email: Brody.Erlandson@colostate.edu
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Shrinkage Priors

Priors:

- $\beta_{j00}^* \sim \text{Normal}(0, 1)$.
- $\beta_{jpd}^* \sim \text{Normal}(0, \sigma_{jp}^2)$ for all p and $d > 0$, where $\sigma_{jp}^2 = \lambda_{jp}^2 \tau_j^2$ and $\lambda_{jp}, \tau_j \sim \text{Half-Cauchy}(0, 1)$ [1].
- $r_{ij} \sim \text{Normal}(0, \phi_j^2)$, where $\phi_j^{-2} \sim \text{Gamma}(a, b)$

Case Study Setup

- We remove the taxa that have fewer than five infants with at least one non-zero count, leaving 16 taxa with a maximum of 98.6% zeros for a taxon.
- We set breast milk proportion $< 10\%$, female, c-section, and multi-patient room as the reference categories.
- Our individual and taxon specific intercept variance hyperparameters (a and b) are set to 3 and 9.
- The at-risk indicators probabilities hyperparameters (α and β) are set to .01 and 10.
- We initialize our coefficients randomly.
- We initialize the at-risk indicators to zero if all counts are zero of a taxon for an infant, and one otherwise.